

Microbiome typing in uveal melanoma is associated with plaque radiotherapy

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ABSTRACT

Background: Microbiomes have been identified in various tumor types and could affect tumor progression and treatment. As the most prevalent primary malignant eye tumor in adults, uveal melanoma (UM) has not been explored regarding its endogenous microbiome. Plaque radiotherapy (PRT) is the gold standard for the treatment of UM. Hereby, we recruited 71 UM patients, sequenced the 16S rRNA gene of their tumor tissues, and analyzed the association between UM microbiome and disease phenotypes.

Results: Clear bacterial signals were observed in UM tissues using fluorescence in situ hybridization. 450 bacterial species passed strict decontamination against 58 environmental control samples in 16S rRNA gene analysis, and these species formed three distinct types by unsupervised clustering. The UM microbiome types were significantly associated with PRT. A biomarker analysis showed that *Pseudomonas* was significantly enriched in the radiation group (RG) compared to the non-radiation group (NRG). A kind of radiation-resistant bacteria had a significantly higher positive rate in tumor tissues that underwent radiotherapy. We found that the radio-resistant bacteria *Deinococcus* was associated with smaller and earlier tumor stages, while *Pseudomonas* and *Stenotrophomonas* were associated with later metastasis.

Conclusion: An endogenous microbiome might exist in UM tissues and was associated with UM features and treatment. Whether the tumor-residing microbiome has a role in UM development and metastasis is worth further investigation.

1. Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults, with an annual incidence in Asian countries of 0.6/1000000 [1,2]. Due to the lack of a lymphatic system in the eye, UM cells enter the systemic circulation mainly through blood vessels, resulting in distant metastasis. Studies have shown that nearly 50% of patients develop metastases, with liver involvement more common, and nearly 80% of patients die within one year after metastasis [3]. Currently, the treatment of UM is mainly divided into enucleation and eye-preserving therapy, in which plaque brachytherapy, a kind of plaque radiotherapy

(PRT), is the main treatment [4].

The human microbiome is made up of microbes that live inside and on the surface of the human body. A core microbiome was reported to share across populations [5] in which the gut microbiome is influenced by differences in diet and host phenotype, thus diversifying the individual microbial ecosystem [6]. The gut microbiome has been shown to influence tumor biology in many ways, such as regulating inflammatory responses, inducing DNA damage, producing metabolites associated with tumorigenesis or tumor suppression, and responding to cancer treatment, including immunotherapy [7]. Previous studies have confirmed that the diversity of intestinal flora in melanoma patients receiving checkpoint

Abbreviations: BRB, Blood-retina barrier; LDA, Linear discriminant analysis; LefSe, Linear discriminant analysis effect size; NRG, Non-radiation group; PRT, Plaque radiotherapy; RG, Radiation group; TSA-FISH, Tyramide signal amplification-fluorescence in situ hybridization; UM, Uveal melanoma.

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blockade immunotherapy is significantly correlated with the therapeutic effect [8], while UM patients do not respond well to checkpoint blockade immunotherapy.

In addition to intestinal flora, bacteria have been found in breast cancer, lung cancer, cutaneous melanoma, and other tumor tissues. And the composition of different types of tumor microbiome is different [9]. The characteristics of tumor flora and metabolic pathways were related to tumor type and subtype, clinical characteristics of patients, tumor development, and therapeutic effect [9,10]. It is speculated that the bacteria selectively colonize tumors with abnormal vascular systems with abundant blood supply and relative leakage, thus growing and multiplying in the relatively hypoxic tumor microenvironment (especially anaerobe or facultative anaerobe). They can regulate enzyme activity, activate specific receptors on cancer cells to make them resistant to chemotherapy, and alter the tumor microenvironment through immune regulation [11]. A recent study found that oral vancomycin was associated with enhanced radiotherapy in cutaneous melanoma xenotransplantation models [12].

Based on the above research background, we speculate that there may be a microbiome in UM. Therefore, 71 cases of UM tissues were sequenced by 16S rRNA and analyzed in the present study, to explore the composition of the UM microbiota and their relationship with PRT and UM phenotypes.

2. Results

2.1. Clinical data of 71 UM samples

A total of seventy-one UM tissues were included in this study and 16S rRNA sequencing was performed, including 36 cases in the radiation group (RG) and 35 cases in the non-radiation group (NRG) (Fig. 1). Our cohort's average age of onset was younger than western patients, NRG was 50.7 ± 11.6 years and RG was 46.7 ± 11.5 years. 34 males and 37 females were included. There were 41 tumors in the right eye and 30 in the left eye. In this study, 7.2% of the tumors were in Stage I, 52.1% were

in Stage II, and 40.6% were in Stage III (The American Joint Committee on Cancer classification, 8th). According to Collaborative Ocular Melanoma Study's recommendation [13], a tumor thickness was no more than 10.0 mm could consider accepting PRT, so there was a significant difference in average tumor thickness between NRG (10.7 ± 3.8 mm) and RG (6.2 ± 2.6 mm) ($p < 0.001$), but there was no significant difference in the average largest basal diameter between the two groups (14.7 ± 4.4 vs. 13.4 ± 3.6 mm).

2.2. Tyramide signal amplification-fluorescence in situ hybridization detected bacterial DNA signals in UM tissues

We performed tyramide signal amplification-fluorescence in situ hybridization (TSA-FISH) in 5 tissues (2 of NRG and 3 of RG) using the probe EUB338 I-III targeting bacterial 16S rRNA. At the same time, we also conducted a negative control experiment and found that the fluorescence signal of specific staining could not be observed without adding EUB probe. Staining for EUB338-probed hybridized sections were detected both in RG and NRG tissues, as shown in Fig. 2. Most bacteria are distributed in the cytoplasm and close to the nucleus. The positive bacterial signals were constantly observed in hybridized sections using the probe EUB338 I-III in all five patients.

2.3. Characterizations of UM microbiome in NRG and RG

A total of 71 tissue, including UM samples (35 of NRG and 36 of RG) and 58 environmental samples (operating room and laboratory environment) have been sent to undergo 16S rRNA gene sequencing (V3–V4 regions). We set four filtering criteria for algorithmic decontamination (threshold score of 0.01, 0.1, 0.5, 0.9, the lower the score, the stricter the decontamination degree) to remove potential contaminations from UM tissue using the forementioned environmental samples as contamination sources. Besides algorithmic decontamination, we also removed four species (*Leptothrix*, *Bradyrhizobium*, *Pedobacter*, and *Variovorax*) that were believed as contaminations in previous studies of intratumoral bacteria.

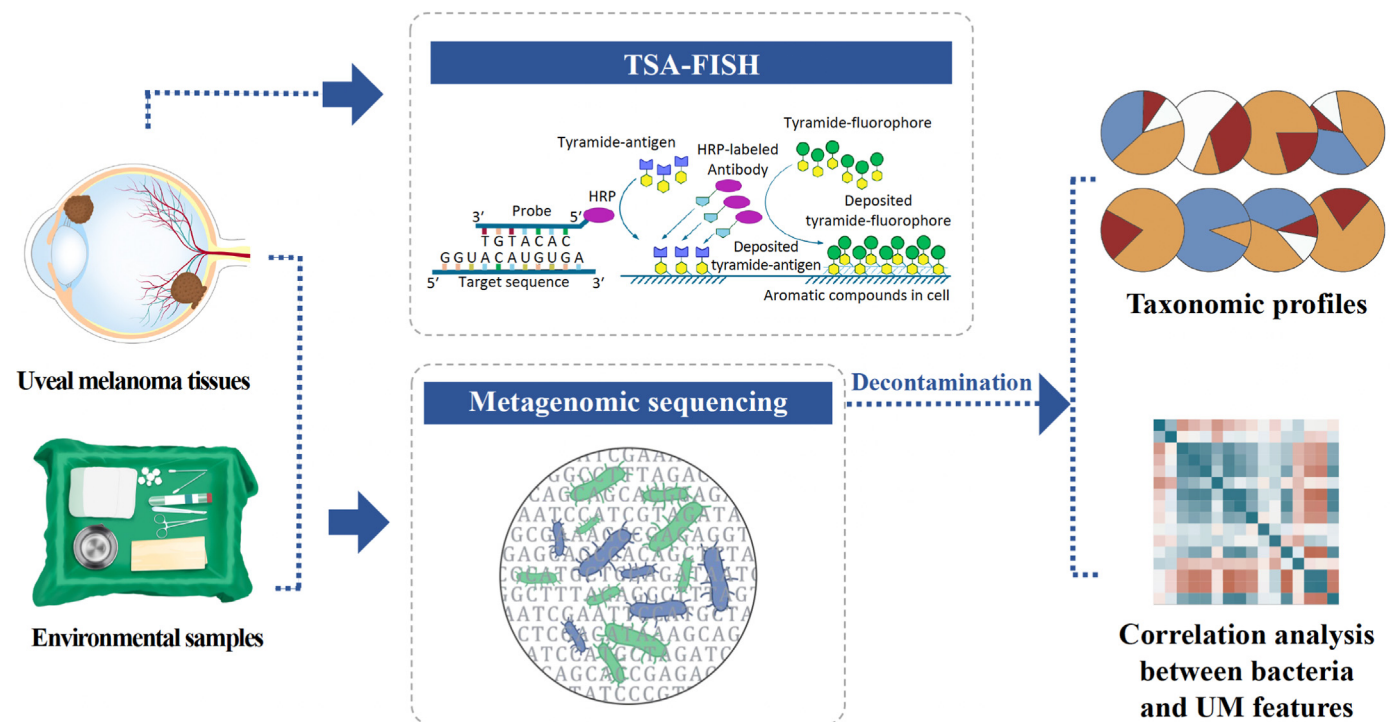


Fig. 1. Overview of the technical route in the current study.

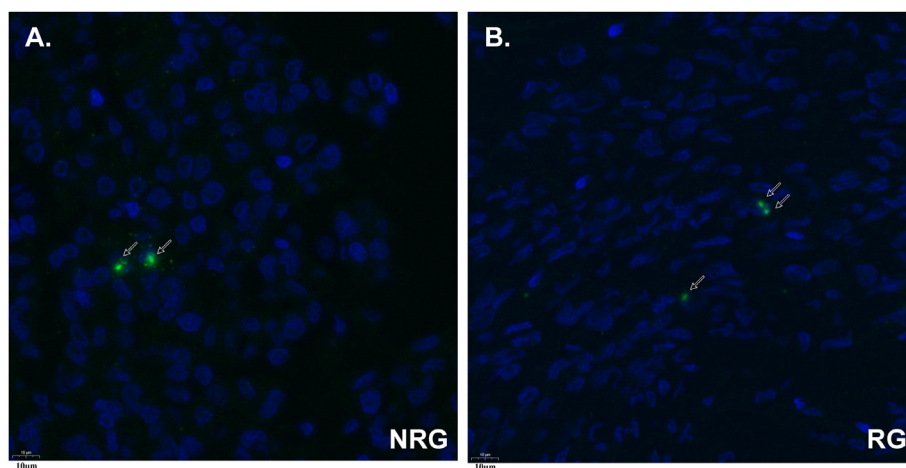


Fig. 2. Visualization of bacteria using TSA-FISH. Representative in situ detection of bacteria obtained by TSA-FISH using DAPI staining (in blue) and HRP-labeled EUB338 I-III probe mixture (in green), showing the presence and distribution of specific hybridized bacterial signals (arrow) in UM tissue. Original magnification: $\times 100.0$; scale bar = 10 μm . A. UM tissue of NRG (n = 2); B. UM tissue of RG (n = 3).

After decontamination, we obtained the OTU number of the corresponding threshold score respectively (Fig. 3A). To ensure the authenticity of the existence of bacteria, we used the OTU table with a decontamination threshold score of 0.01, removed the samples with reads less than 2000 (NR34, R21, and R26) and standardized analysis according to the minimum sequencing quantity of the remaining samples, which OTU number was 450 and had 144,092 reads as the basis for further analysis.

We hope to observe the OTU number and relative abundance of all kinds of bacteria in NRG and RG at different species classification levels. The results showed that the UM tissue was dominated by *Proteobacteria* at the phylum level, while in some RG samples, the abundance of *Firmicutes* and *Bacteroidetes* was relatively high (Fig. 3B). At the family and genus level, the abundance of different species in each sample was relatively stable (Fig. 3C and D). The relative abundance of *Stenotrophomonas* in 12 RG samples was more than 50%, while only 6 in NRG. We found only 12 species whose relative abundance was greater than 0.01 and can be classified at the genus level. They were *Stenotrophomonas*, *Enhydrobacter*, *Aerococcus*, *Acinetobacter*, *Micrococcus*, *Elizabethkingia*, *Deinococcus*, *Comamonas*, *Sphingomonas*, *Pseudomonas*, *Methylobacterium*, and *Brevundimonas*. The results showed that the main component of UM microbiome is *Proteobacteria*, including *Stenotrophomonas* and *Enhydrobacter*.

2.4. Microbiota types in UM tissues

We referred to the enterotype classification method of the gut microbiota [14] and performed it in UM samples. The UM samples formed three clusters that we designate them as UM 'Types' (Fig. 4, Supplement Figure 1). Type 2 markers were *Stenotrophomonas*, *Sphingomonas*, *Pseudomonas*, *Methylobacterium*, and *Elizabethkingia* (Supplement Fig. 1A-E), the enrichment of these bacteria in Type 2 was significantly higher than that of Type 1 and Type 3. Type 3 markers were *Acinetobacter* and *Micrococcus* (Supplement Fig. 1H and I). Surprisingly, *Enhydrobacter* and *Deinococcus* were not only enrichment in Type 3 but also significantly different among three types. Although there was no statistical difference between NRG and RG in the classification of microbiota types, we can find that there were more NRG samples in Type 1 and more RG samples in the other two types (Fig. 4A). Next, we compared the relative abundance of 12 species and 'others' in RG and NRG. We found that the relative abundance of *Pseudomonas*, which was enriched in Type 2, was also higher in RG (Supplement Figure 2A), and the number of samples in RG in Type 2 is more than that in NRG (Fig. 4A).

2.5. Differential expression and correlation network of bacteria in NRG and RG

Under the strictest threshold (0.01) of decontamination, a comparison of the alpha diversity indices showed no significance between NRG and RG. Additionally, linear discriminant analysis effect size (LEfSe) analysis showed that *Pseudomonas* was significantly higher in RG than that in NRG (linear discriminant analysis (LDA) score >2.0 , $p < 0.05$) (Fig. 4C). We further investigated the correlation of bacteria in RG and NRG at the genus level. The inclusion criteria with a correlation coefficient over 0.4 were selected and presented in the form of the network diagram in Fig. 4D. Although there was only positive correlations among all species except 'others' in the two groups, it was still found that the bacteria were divided into two clusters in the RG. In the correlation network, *Stenotrophomonas*, with the highest abundance, has a positive correlation with only one cluster (left cluster).

2.6. Correlating the UM microbiota composition with clinical characteristics

In order to investigate possible correlations between the microbiota composition of UM tissues and clinical characteristics, we included clinical features that may be related to the severity of the disease, calculated their associations with different bacteria, and inferred whether a species is related to the progression of the disease (Fig. 5, Table 1). We observed that *Aerococcus* and *Deinococcus* were correlated with smaller and earlier tumors (adjust $p < 0.05$), which suggested that they may provide protection in the early stage of tumor. In analyses related to clinical outcomes, *Pseudomonas* and *Stenotrophomonas* were associated with later metastasis (adjust $p < 0.05$). More basic experiments and clinical trials will be needed to verify our results.

3. Discussion

With the improvement of microbial detection technology, scientists have paid more and more attention to the research of flora. In recent years, variates studies have found that the existence of bacteria in tumors and were related to the occurrence and development of disease [9,15]. In this study, we found the existence of bacteria in UM for the first time, explained the characteristics of UM microbiota and its relationship with PRT and clinical features. It remains controversial whether bacteria are present in tissues previously thought sterile. The first conceptual evidence for bacterial translocation was reported in 2011 when Amar et al.

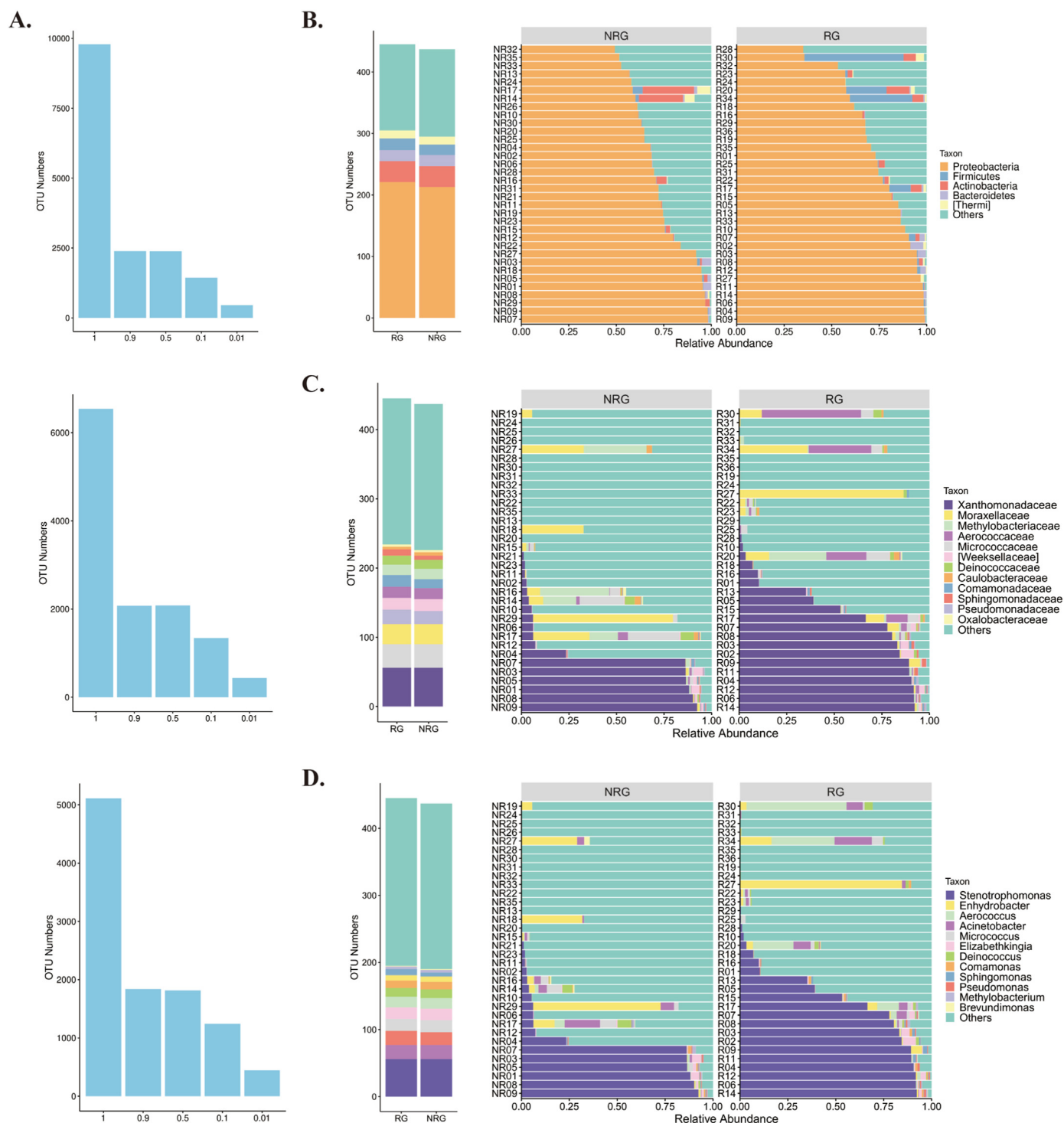


Fig. 3. Effect of algorithmic decontamination and composition of UM microbiome. A. OTU number histogram with different degrees of decontamination. Histogram of All samples (top), NRG (median) and RG (bottom). B-D. OTU number (left) and relative abundance (right) of all kinds of bacteria in NRG and RG at different species classification levels. B. Phylum level of species. C. Family level of species. D. Genus level of species.

showed that bacterial 16S rDNA gene was detectable in blood [16]. After tumorigenesis, the destruction of physical and molecular barriers, coupled with relative immunosuppression in the tumor microenvironment, may increase the likelihood of bacterial translocation to normally aseptic sites. Then various studies supported the hypothesis of translocation bacteria [17,18]. A recent study found that bacteria are widespread in different types of tumors (including exposed and protected), of which cutaneous melanoma has the lowest number of bacteria [9]. We

first presented the composition of the human UM microbiota and found that it could be divided into three types in the present study. We also observed differences in UM microbiota between NRG and RG and evidenced that bacterial signatures were significantly associated with UM clinical characteristics.

Even though we use strict aseptic procedures during the operation, we still collect a large number of environmental control samples for algorithmic decontamination. After using the most stringent decontamination

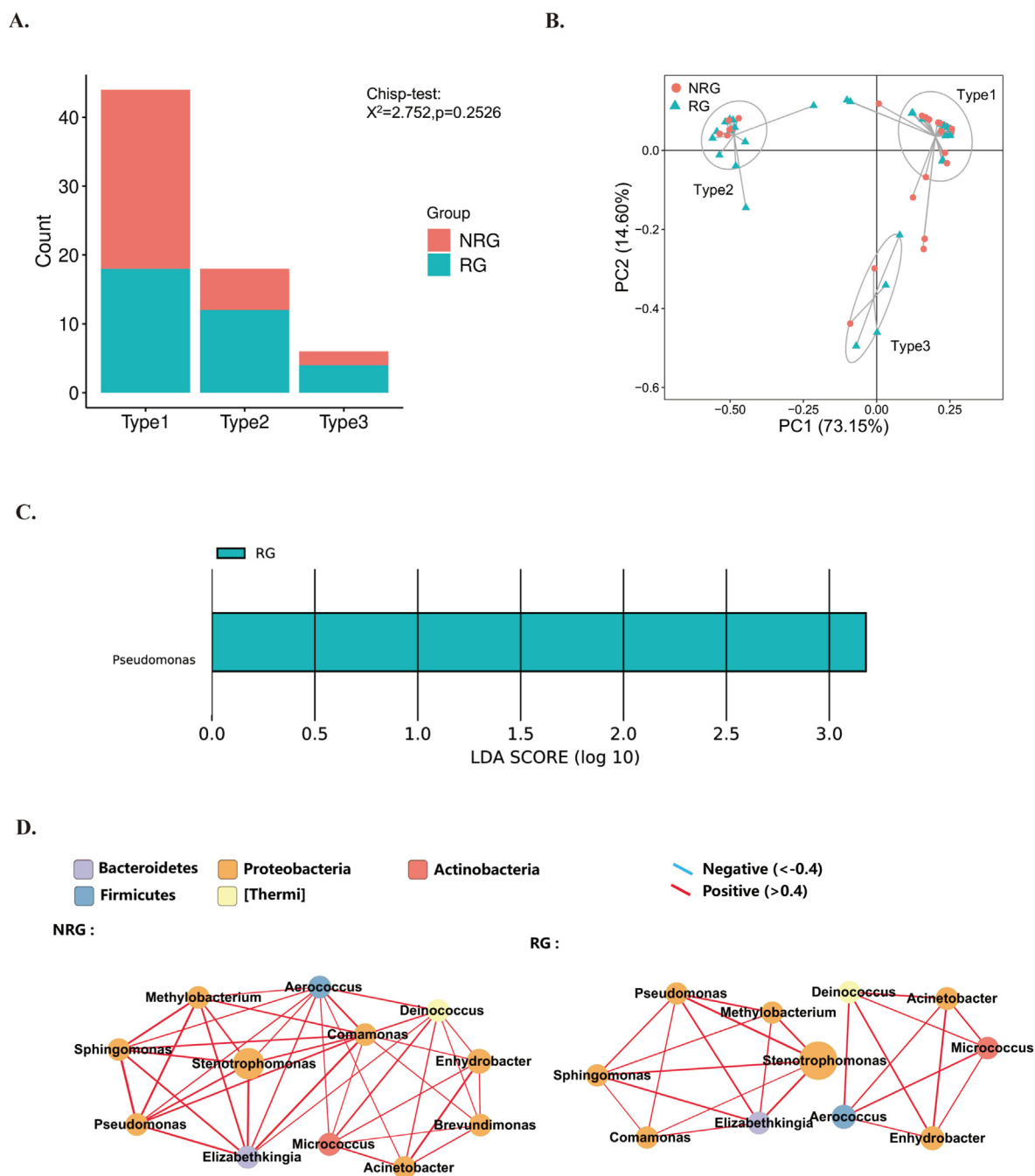


Fig. 4. Different UM types and differential expression and correlation network of bacteria in two groups.

A. Composition of RG and NRG samples in each type. B. Types of UM samples were visualized by PCoA and clustering at the genus level. C. Significant different bacteria enrichment between NRG and RG. The results could be biomarkers in RG. LDA score >2.0 , $p < 0.05$. D. Co-occurrence networks of NRG and RG at the genus level. The size of the circle represents the abundance of bacteria. The thickness of the connected lines represents the strength of the relationship between bacteria.

threshold of 0.01, we removed about 95% of the species. After removing four potential contamination bacteria, there were still real bacterial signals, so the conclusion that bacteria were found in UM tissues in this research is credible. This study detected positive bacterial signals in both NRG and RG UM samples using TSA-FISH, distributing in cell plasma but not in nuclear.

After 16S rRNA sequencing and OTU annotation, we found that the UM microbiota was mainly *Proteobacteria*, and at the genus level, it was mainly *Stenotrophomonas*, *Acinetobacter*, and *Pseudomonas*. In the recent report which studied on bacteria in the eye was found that the characteristics of microbiota in the aqueous humor of different eye diseases were different, but *Proteobacteria* was also dominant [19]. *Stenotrophomonas* has the highest relative abundance in UM microbiome,

which is consistent with a recent report about hepatocellular carcinoma [15]. Previous studies have shown that as a new opportunistic pathogen, patients with long-term neutropenia and exposure to broad-spectrum antibiotics have a higher risk of *Stenotrophomonas* infected, leading to pneumonia, bacteremia, catheter-related infection, and other diseases [20]. Its pathogenic conditions in the eyes are mostly bacterial keratitis [21] (caused by trauma, incorrect wearing of corneal contact lens [22], etc.) and infectious endophthalmitis after cataract surgery [23]. *Stenotrophomonas* can not only secrete adventitial vesicles and elicit a strong inflammatory response [24], but also have been found to be positively correlated with regulatory T cells that inhibit anti-tumor immune response in gastric cancer tissues and participate in the formation of the

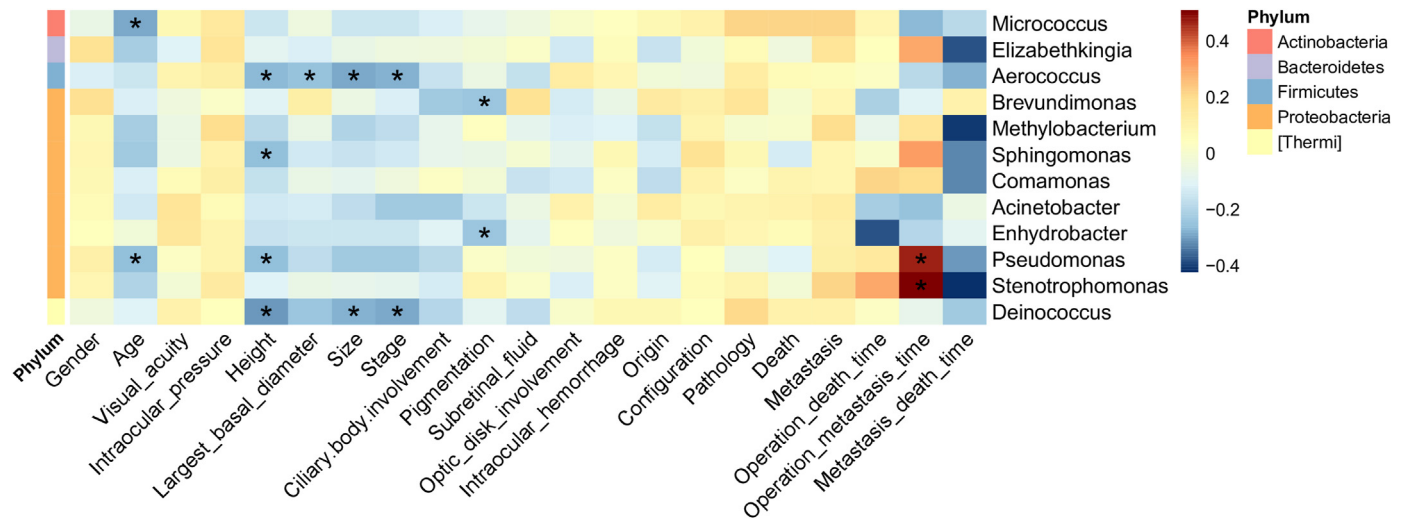


Fig. 5. Heatmap of bacteria and clinical characteristics. Cross axis listed clinical characteristics. The vertical axis showed different bacteria at the genus level. *: adjust $p \leq 0.05$. ***: adjust $p \leq 0.001$.

Table 1

Clinical prognosis characteristic of NRG (n = 35) and RG (n = 36).

Clinical prognosis characteristic	NRG	RG	P-value	Hazard Ratio
Operation-Metastasis Time (months)	14.8 ± 21.3	51.7 ± 26.5	0.393	0.572
Operation-Death Time (months)	25.8 ± 19.0	45.5 ± 17.1	0.370	0.593
Metastasis-Death Time (months)	12.0 ± 8.7	7.0 ± 5.8	0.632	1.371

immunosuppressive microenvironment of gastric cancer [25]. *Stenotrophomonas* induces NLRP3 inflammasome complex formation through the toll-like receptor (TLR) 4-mediated NF- κ B signaling pathway and secretes various inflammatory factors in the liver, thereby promoting the progression of hepatocellular carcinoma [15]. TLR is a class of receptors that can recognize the components related to various pathogens and destroy them by inducing immune and inflammatory responses, playing an essential role in the prevention of infectious diseases. Lipopolysaccharide is a structural component on the cell wall of Gram-negative bacteria and is the most important activator of TLR4. However, improper activation of TLRs can lead to an uncontrolled inflammatory response that can lead to disease. Recent studies have found that lipopolysaccharide can increase TLR4 mRNA and protein of uveal melanocytes, which may have a protective effect on the eye. However, excessive lipopolysaccharide-mediated inflammation may also lead to the progression of uveal inflammation [26]. A high positive rate of lipopolysaccharide was also found in cutaneous melanoma tissues [9]. These results suggested that *Stenotrophomonas* may play a key role in the occurrence and development of UM by stimulating uveal melanocytes to produce more TLR4 protein through lipopolysaccharide, activating NF- κ B signaling pathway, resulting in an abnormal inflammatory response, or changing tumor microenvironment. Of course, this hypothesis will require extensive experiments and analysis at the molecular and cellular levels. Otherwise, there were more annotated species in RG with relative abundance >0.01 than in NRG. While the proportion of 'others' in some samples was very high, which may be due to the following reasons: First, the species diversity was high, but the relative abundance was less than 0.01. Second, some species cannot be classified to the level of family or genus. Third, there may be new species that have not been discovered before.

A very interesting phenomenon is that all the UM samples formed three distinct clusters, which was in accordance with studies in gut

microbiota enterotypes [14]. Among them, RG samples accounted for more in Type 2 and Type 3. To explore the microbiome characteristics of different types, we compared the relative abundance of all detected bacteria in the three types and found that *Stenotrophomonas* and *Pseudomonas* were enriched in Type 2 and had statistical significance (Supplement Figure 1, $p < 0.0001$). In the correlation network of bacteria in RG (Fig. 4D), the correlation between *Stenotrophomonas* and *Pseudomonas* were stronger than other species. A recent skin microbiota study of patients with radiation dermatitis after radiation therapy found that the co-existence of *Stenotrophomonas* and *Pseudomonas* was significantly associated with delayed healing of radiation dermatitis [27]. This is consistent with the previously reported results of interaction between *Stenotrophomonas* and *Pseudomonas* in pulmonary cystic fibrosis. *Stenotrophomonas* may give a selective "fitness advantage" to *Pseudomonas* in chronically infected environment or worsen pulmonary disease by increasing the virulence of the *Pseudomonas* [28]. Among patients diagnosed with chronic sinusitis after nasal and skull base radiotherapy, *Pseudomonas* was the second most common bacterial species in the sinusitis [29], and *Pseudomonas* was also isolated from the blood of patients receiving radiotherapy for oral squamous cell carcinoma [30]. Therefore, the interaction between *Stenotrophomonas* and *Pseudomonas* and the relationship between these two species are worthy of further study in UM.

To explore the effect of PRT on UM microbiota, we analyzed RG and NRG by LEfSe analysis to find out the difference between the two groups. The results showed that the abundance of *Pseudomonas* was higher in RG. Previous research showed that *Pseudomonas aeruginosa* ExoT induces G1 cell cycle arrest in melanoma cells, which significantly inhibits the growth of melanoma [31]. Lipopolysaccharide secreted by *Pseudomonas solanacearum* had an obvious anti-metastasis effect in melanoma B-16 experimental animals [32]. *Pseudomonas aeruginosa* was found to be the second most common bacterial species in nasopharyngeal necrotic secretions of patients with nasopharyngeal cancer after radiotherapy, and the cure rate of patients with negative pathogen culture was significantly higher than that of patients with positive pathogen culture [33]. We found that there was different in the positive rate of *Deinococcus* with strong radiation resistance [34] between RG (16 of 34 cases) and NRG (9 of 34 cases) ($p = 0.07$) after standardized. Although there was non-significant, we still speculate that PRT influences UM microbiome.

We analyzed the correlation between 12 species and clinical information to explore the influence of UM microbiota on clinical characteristics and disease prognosis. *Deinococcus radiodurans* can survive in various environments with extreme radioactive and ultraviolet radiation

[35], and its extract Deinoxanthin can induce apoptosis of cancer cells [36]. This study found that *Deinococcus* was associated with small and early tumors, while the positive rate of *Deinococcus* in RG was high, and the metastasis and death hazard in patients with RG was significantly lower than that in NRG (with non-significant) (Table 1), suggesting that *Deinococcus* may play a role in inhibiting tumor development. *Stenotrophomonas* was found to be associated with a better prognosis in this study, but it was thought in previous studies that it could elicit a strong inflammatory response [24] and enrich in tumor tissue [37], which is inconsistent with our conclusion and needs further experimental verification. *Aerococcus* was considered to be a common pathogen of urinary tract infection, and it has been found enriched in the cervicovaginal microbiota of cervical squamous cell carcinoma in a previous study [38]. Kalker et al. have found that the abundance of *Aerococcus* in gut microbiome decreased significantly after radiotherapy [39]. But in our research, *Aerococcus* was correlated with small and early tumors, and the relative abundance was no significant differences between RG and NRG.

Although the eye can also be connected with the nasal sinus or conjunctiva, due to the anatomical position of the choroid (which is the middle layer of the wall of the eyeball), we do not give priority to the possibility of bacterial translocation from adjacent parts. The blood-retina barrier (BRB) is the basis of immune privilege in eye, which is divided into outer BRB (choroidal vascular endothelial cells) and inner BRB (tight junctions between RPE cells). However, the integrity and permeability of BRB are dynamic, which means that BRB may be affected by different factors [40]. Pathogens in the bloodstream may penetrate the BRB by three ways: transcellular route, Trojan horse (infected leukocyte), or paracellular route [41]. Recent research found that BRB cells may protect the neuronal retina from blood-original pathogens by producing antimicrobial peptides such as lysozyme [42]. Therefore, we speculate that the bacteria in UM are probably bacteria from other body parts that transfer to the choroid through the blood. However, all the above hypotheses need to be proved by experiment data.

There were still some limitations in this study. First, the tumor tissue is small and lacks adjacent tissue samples. Due to the small size of the eyeball, the limitation of local excision, and the improvement of the eye preservation rate of UM, we can rarely obtain tumor and adjacent samples, especially the normal control samples. Second, lack of detection of living bacteria and metabolic function. Although we have detected bacteria in 16S rRNA and TSA-FISH, we cannot directly observe bacteria. We can add electron microscope detection in future study, supplemented with living bacterial biomarkers and bacterial culture to verify the existence of living bacteria, which is a great challenge, because it requires stringent experimental conditions and techniques. Third, we cannot determine the source of the bacterial DNA detected in UM. Whether the bacteria were derived from the gut or adjacent sites, we need more types of samples and experiments for further exploration. Last, the analysis of UM tissues was based on the strictest threshold of decontamination, which actually may lose real bacterial signals. However, how to set the decontamination threshold to maintain the authenticity of the sequencing results best and eliminate the interference of environmental contamination is still controversial. Nevertheless, the previous study has shown that the number of bacteria and Shannon Index in cutaneous melanoma were also the lowest compared to other types of tumors [9]. Although we found that the bacterial load is relatively low, but our sequencing depth is enough (Supplement Figure 3). Therefore, more large-scale basic research and bioinformatics analysis are needed to explore whether the UM microbiome plays a causal role in UM occurrence, treatment, and progression.

4. Conclusions

In conclusion, our study provides evidence for the existence and composition of bacteria in UM, exploring the relationship between UM's microbiome and its clinical characteristics for the first time. There are

differences in tumor microbiota in UM patients treated with PRT or not, and UM could be divided into three types according to the microbiome, which is associated with PRT.

5. Methods

We chose the tumor tissues of 71 patients treated in Beijing Tongren Hospital. All samples were obtained by local resection of the tumor or enucleation. This study was in accordance with the Helsinki Declaration and approved by the Ethics Committee of Beijing Tongren Hospital, Capital Medical University (Ethics Approval No.: TRECKY2018-056). Every patient signed an informed consent to participate in the study. We divided 71 tumor samples into two groups according to whether they received PRT (RG) or not (NRG), and 16S rRNA sequencing and analysis were carried out. In addition, we selected UM tissues from 5 patients for TSA-FISH, including 3 cases of RG and 2 cases of NRG. All samples were collected under aseptic conditions during the operation. At the same time, 58 environmental negative control samples were collected, including operating room air, sterile clothes, dry gauzes, wet gauzes, instrument washing solution, sterilized skin swabs, physiological saline, sterile towel swabs, sampling tubes, aseptic buffer solutions, laboratory air, sterile water and other environmental samples. The specimens were transferred directly from the operating room to the biosafety cabinet in the aseptic container in 20 min. All samples were stored at -80°C until undergoing TSA-FISH or 16S rRNA sequencing. The main diagnostic and treatment criteria are consistent with the Collaborative Ocular Melanoma Study [43]. We recorded demographic information (age and sex), eye and tumor features (best-corrected visual acuity, intraocular pressure, laterality, tumor size, shape, pigmentation, location, optic disc involvement, association with subretinal fluid, intraocular hemorrhage, ciliary body involvement, and pathological types). Whether or not to receive PRT depends on the characteristics of the tumor [4] and the patient's will. Otherwise, we recorded the time of tumor metastasis and death.

5.1. TSA-FISH

TSA-FISH was performed following a modified protocol described by Choi et al. [44]. The basic steps included paraffin section dewaxing and tissue rehydrating, pretreatments of tissue sections, hybridizing probe, and detection with anti-digoxigenin/horseradish peroxidase antibodies (anti-DIG HRP-conjugated antibody). EUB338 I-III were specific bacterial probes that could bind to the 16S rRNA region of most bacteria. Digoxin labeled EUB probe was selected to label bacteria in UM tissues. Anti-digoxin/horseradish peroxidase antibody was used as a secondary antibody to bind to EUB probe and was observed under the same fluorescence microscope after amplifying Tyramine signal. All steps were performed under sterile conditions.

5.2. 16S rRNA gene amplification and sequencing

Microbial DNA was extracted from UM samples (36 of RG and 35 of NRG) and negative samples (58 of environmental samples) using a DNA extraction kit according to the manufacturer's protocols. The concentration and purity were measured using the NanoDrop One (Thermo Fisher Scientific, MA, USA). Next, 16S rRNA genes of V3–V4 hypervariable regions were amplified with the primers 338F (5'-ACTCTACGGGAGG-CAGCA-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') using the thermocycler PCR instrument BioRad S1000 (Bio-Rad Laboratory, CA, USA). The length and concentration of the PCR product were detected by 1% agarose gel electrophoresis. Samples with a bright main strip between could be used for further experiments. PCR products were mixed in equidensity ratios according to the GeneTools Analysis Software (V4.03.05.0, SynGene). Then, the mixture of PCR products was purified with E.Z.N.A. Gel Extraction Kit (Omega, USA). After the purification of PCR products, sequencing libraries were generated using NEBNext® Ultra™ DNA Library Prep Kit for Illumina® (New England Biolabs, MA,

USA). The library quality was assessed on the Qubit@ 2.0 Fluorometer (Thermo Fisher Scientific, MA, USA) and sequenced on Illumina Novaseq 6000 System with Mode PE250, and 250 bp paired-end reads were generated.

5.3. Decontamination and data analysis

All 16S rRNA gene datasets were processed using a standardized pipeline in QIIME 2 [45]. For each dataset, demultiplexed sequencing reads from the UM tissues and environmental control samples were denoised to generate amplicon sequence variants (ASVs) using the software DADA2 (V1.6) [46].

The Bayesian approach based SourceTracker (V0.9.1) [47] was used to estimate source proportions in UM samples that came from the environment according to the species community structure of Sink (tumor samples) and Source (environmental samples). Another decontaminative software Decontam (V1.10.0) (<https://github.com/benjineb/decontam>), an open-source R package that implements a statistical classification procedure that identifies contaminants in 16S data, was used to filter out features on genus level with a high chance of being contaminants using sequenced environmental control samples [48]. Prevalence-based contaminant identification was used with four different threshold scores of 0.01, 0.1, 0.5, and 0.9, respectively. In addition to the algorithm decontamination, after referring to a large number of studies on bacteria in tumors [19,49–55], we believe that *Leptothrix*, *Bradyrhizobium*, *Pedobacter*, and *Variovorax* may also be the potential contaminant species. We removed mitochondria, chloroplasts, unclassified species, four potential bacteria, and the samples with reads less than 2000, then standardized analysis according to the minimum sequencing quantity of the remaining samples. After combining the above-mentioned methods, we obtained four OTU tables with different decontaminative levels.

Clustering and classification were performed using Jensen–Shannon distance and partitioning around medoid (PAM) clustering. The optimal number of clusters was estimated using the Calinski–Harabasz index [14].

Alpha diversity was applied in analyzing the complexity of species diversity for a sample through four indices, including Observed species, Chao 1, Shannon, and Ace, based on final OTU tables. All these indices in our samples were calculated with QIIME 2 and displayed with R software (V2.15.3).

Beta diversity analysis was used to evaluate differences of samples in species complexity. Principal Coordinate Analysis (PCoA) was performed to get principal coordinates and visualize from complex, multidimensional data. PCoA analysis was displayed by QIIME 2 and ggplot 2 package in R software. NMDS analysis was performed by the vegan package of R software based on the weighted and unweighted unfract distance matrix.

LEfSe analysis was used to find the biomarker of NRG and RG based on the final OTU table with the threshold score of 0.01. Non-parametric factorial Kruskal–Wallis (KW) sum-rank test detected species with significant differences in abundance between different groups, and Wilcoxon rank sum test was used to judge the difference between the two groups. Finally, LDA was used to achieve dimensionality reduction and evaluate the impact of significantly different species by setting LDA score ≥ 2 and obtained the biomarkers in different groups.

Correlation analysis and variation partitioning analysis were performed by R software based on the key genus (relative abundance $\geq 1\%$), and canonical correspondence analysis was analyzed based on the final OTU table with a threshold score of 0.01. Spearman correlation index between species or samples based on the OTU table was calculated by R software and visualized by Cytoscape (<http://cytoscape.org/>).

Correlations of the taxonomy counts (phylum and genus relative abundances) and disease features, such as demographic information, clinical features, and follow-up results were estimated using the Spearman correlation coefficient and represented in a correlation plot using the R Corrplot package.

5.4. Statistical analysis

The statistical analyses were performed using the Statistical Product and Service Solutions (SPSS) software version 25 (International Business Machines Corporation, Armonk, New York, United States). Students' t-test was used to determine the difference between the two groups. The positive rate of *Deinococcus* in RG and NRG was analyzed by chi-square test. Statistical significance was considered when p-value was less than 0.05. Cox multivariate regression analysis was used to evaluate the difference in clinical prognosis between RG and NRG.

Ethics approval and consent to participate

This research adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board of Beijing Tongren Hospital. Furthermore, the study obtained the informed consent of all patients.

Availability of data and materials

Raw data of all samples were successfully uploaded on NCBI and the data will be released at <https://www.ncbi.nlm.nih.gov/sra/PRJNA807130> to the public on June. 30th, 2023.

Author contribution

Yuning Chen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Writing - Original Draft, Project administration. Jingting Luo: Conceptualization, Methodology, Investigation, Resources, Writing - Review & Editing. Rui Fang: Conceptualization, Resources. Ruiheng Zhang: Conceptualization, Resources. Yueming Liu: Resources. Haowen Li: Investigation. Jingying Xiu: Data Curation. Jie Xu: Visualization. Yang Li: Project administration, Funding acquisition. Wenbin Wei: Supervision, Project administration, Funding acquisition. All authors read and approved the final manuscript.

Conflict of Interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.medmic.2023.100079>.

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